

Scalability & Future Plan

Date	10/07/2026
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Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current **Credit Card Approval Prediction** system can be enhanced to support larger datasets, increased user traffic, improved prediction accuracy, and enterprise-level deployment. The project currently uses a machine learning model integrated with a Flask web application to predict whether a credit card application should be approved or rejected. Future enhancements focus on improving scalability, security, performance, and integration with real-world banking systems.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited dataset used for model training (around 200+ records).	May reduce prediction accuracy for diverse real-world applicants.	High
2	Flask application is designed for single-user/local deployment.	Cannot efficiently support multiple concurrent users.	Medium
3	No authentication, database, or external banking API integration.	Limits enterprise deployment and secure user management.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports individual/local usage through Flask.	Deploy using cloud platforms (AWS, Azure, or GCP) with Gunicorn/Nginx for handling multiple concurrent users.
2	Data Storage	Applicant data stored in CSV files.	Use MySQL/PostgreSQL or MongoDB for secure and scalable storage.
3	Performance	Prediction executed on a locally trained model.	Optimize preprocessing, enable model caching, and deploy using REST APIs for faster response.

4	Security	Basic input validation without authentication.	Implement user authentication, HTTPS, JWT security, encrypted data storage, and role-based access control.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve model accuracy using larger datasets and advanced ML algorithms (XGBoost, LightGBM).	1–2 Months	Higher prediction accuracy and better generalization.
Phase 2	Integrate relational database and user authentication system.	2–3 Months	Secure data management and support for multiple users.
Phase 3	Deploy the application on cloud platforms with REST API support and real-time monitoring.	3–4 Months	High availability, scalability, and enterprise-ready deployment.
Phase 4	Integrate credit bureau APIs, explainable AI (XAI), and mobile application support.	4–6 Months	Improved decision transparency, enhanced user experience, and real-world banking integration.